

Unconditional interval-certified compact-support Weil positivity at $L = 0.3$

via the x -space kernel form

PRIME.RDAGGER.KERNEL_LOEWNER.01 — round 494

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Abstract

The three-gate contract of the round-492 judgment is executed in its prescribed order. For every real $h \in L^2(\mathbb{R})$ supported in $[-0.3, 0.3]$,

$$Q_W(h) \geq 2.1 \cdot 10^{-3} \|h\|_2^2.$$

The proof is unconditional and interval-certified. It uses the classical x -space Weil kernel, positivity of every compressed translation difference, a sign-safe cutoff, and one 401-dimensional Legendre finite section with an enclosed exact Hilbert–Schmidt rest. The Galerkin values of the uncut form are calibration upper bounds only; none is used as an operator lower bound. This is an experiments-side compact-support theorem. It is not a claim for or against the Riemann Hypothesis.

Status. PROVED($c=2.1000e-3$). No ledger row and no promotion of an RH claim.

Probe SPEC_SHA: 2e6d71138dff454025641443a9c77f8018d36827dcf94cc35f1696dae9deda47.

Claim boundary. The theorem is a fixed-support statement about the Weil quadratic form at $L = 0.3$. It proves neither RH nor anti-RH and contains no claim about zeros. It does not assert that a finite Galerkin eigenvalue is the operator eigenvalue. The operator floor comes from a Loewner minorant plus a complete finite-section remainder bound.

1 The theorem

Let $\tilde{h}(x) = h(-x)$ and $g = h * \tilde{h}$. For real h put

$$\Pi(h) = 2 \left(\langle h, \cosh(\cdot/2) \rangle^2 - \langle h, \sinh(\cdot/2) \rangle^2 \right).$$

Theorem 1 (Unconditional interval-certified compact-support Weil positivity at $L = 0.3$ via the x -space kernel form). *If $\text{supp } h \subset [-0.3, 0.3]$, then*

$$Q_W(h) \geq 2.1 \cdot 10^{-3} \|h\|_2^2.$$

The same inequality holds for complex h after applying the real statement to its real and imaginary parts.

Since $2L = 0.6 < \log 2$, the prime sum is empty. Thus this theorem is scramble-insensitive only vacuously; no source positions occur at this support length.

2 Gate G1: the sealed kernel identity

Lemma 1 (x -space Weil form). For $2L < \log 2$,

$$Q_W(h) = \int_0^{2L} k(x)(g(0) - g(x)) dx - c_L g(0) + \Pi(h), \quad (1)$$

$$k(x) = \frac{e^{x/2}}{\sinh x}, \quad (2)$$

$$c_L = \int_0^{2L} \left(k(x) - \frac{1}{x} \right) dx + \log(4L) + \gamma + \log \pi. \quad (3)$$

At $L = 0.3$,

$$c_L \in [2.19240491113, 2.19240491114], \quad c_L = 2.1924049111319993023 \dots$$

Proof. For the defining archimedean multiplier, Gauss' integral for $\psi(\frac{1}{4} + it/2)$ and Plancherel give, after $v = 2x$,

$$A(h) = \int_0^\infty \left(\frac{e^{-2x}}{x} g(0) - k(x)g(x) \right) dx - (\log \pi)g(0).$$

The autocorrelation vanishes for $x > 2L$. Splitting there and using

$$\lim_{\varepsilon \downarrow 0} \left(\int_\varepsilon^{2L} \frac{dx}{x} - \int_\varepsilon^\infty \frac{e^{-2x}}{x} dx \right) = \log(4L) + \gamma$$

proves the archimedean part of (1). Furthermore $\widehat{g}(i/2) + \widehat{g}(-i/2) = 2(\langle h, \cosh \rangle^2 - \langle h, \sinh \rangle^2)$, which fixes the pole sign, including the negative odd contribution. The prime term is zero because $e^{2L} < 2$. $\square$

The three mandatory stop hits. The x -side regularization

$$\int_0^\infty \left[\frac{e^{-2x} - 1}{x} - \left(k(x) - \frac{1}{x} \right) \right] dx - \log \pi$$

gives

$$\sigma_A(0) = -5.3721834192256654.$$

Its comparison with $\psi(\frac{1}{4}) - \log \pi$ differs by $3.503 \cdot 10^{-26}$ after cutting the exponentially small tail at $x = 120$. For $h(x) = \cos(\pi x/(2L))$, direct insertion in (1) gives

$$\frac{Q_W(h)}{\|h\|^2} = 1.18025311091317 \cdot 10^{-2} \in [1.150, 1.192] \cdot 10^{-2}.$$

The odd Legendre section gives $2.2256359052 \cdot 10^{-1}$, consistent with the sealed $0.211/0.221/0.226$ ladder. Here $\Pi_{\text{odd}} = -2|\sinh(\cdot/2)\rangle\langle\sinh(\cdot/2)|$; the opposite sign is not used. All three stop conditions pass.

3 Gate G2: Loewner truncation with boundary strips

Let $E : L^2[-L, L] \rightarrow L^2(\mathbb{R})$ be extension by zero and let τ_x be the unitary translation $(\tau_x f)(y) = f(y+x)$ on $L^2(\mathbb{R})$. Its interval compression is $R_x = E^* \tau_x E$. For $0 \leq x \leq 2L$,

$$\begin{aligned} 2\langle h, (I - \text{Re } R_x)h \rangle &= \|Eh - \tau_x Eh\|_{L^2(\mathbb{R})}^2 \\ &= \int_{-L}^{L-x} |h(y) - h(y+x)|^2 dy \\ &\quad + \int_{-L}^{-L+x} |h(y)|^2 dy + \int_{L-x}^L |h(y)|^2 dy \geq 0. \end{aligned}$$

The last two integrals are the translation boundary strips. They would be lost if the norm were incorrectly taken only on $[-L, L]$. Consequently

$$I - \operatorname{Re} R_x \succeq 0 \quad \text{for every } x,$$

and multiplication of the x -integral by any $0 \leq w \leq 1$ gives a Loewner minorant.

The sealed cutoff is

$$w(x) = \begin{cases} 0, & 0 \leq x \leq x_0, \\ 6t^5 - 15t^4 + 10t^3, & x_0 < x < x_1, \quad t = (x - x_0)/(x_1 - x_0), \\ 1, & x_1 \leq x \leq 2L, \end{cases} \quad (x_0, x_1) = (0.01, 0.03).$$

This specific ramp is C^2 (not C^∞); C^2 is sufficient for the Hilbert–Schmidt certificate below. Split 80-digit quadrature gives

$$\kappa_w = \int_0^{0.6} w(x)k(x) dx = 3.6980080402989774725 \dots \in [3.69800804029, 3.69800804031].$$

Define T_w by the kernel

$$T_w(y, z) = \frac{1}{2} w(|y - z|) k(|y - z|).$$

Then

$$Q_W \succeq Q_W^{(w)} = (\kappa_w - c_L)I - T_w + \Pi.$$

The omitted interval is therefore booked by using κ_w , never the uncut diagonal coefficient. As an audit diagnostic, the uncut $N = 36$ Galerkin value $7.5719610133 \cdot 10^{-3}$ falls to the truncated finite target $4.2375222058 \cdot 10^{-3}$, a booked loss $3.3344388074 \cdot 10^{-3}$, below half of the uncut diagnostic margin. This comparison is not an operator bound and is not used in the proof.

4 Gate G3: finite section and exact rest

Let P_n project onto the normalized Legendre polynomials of degrees $0, \dots, n-1$ on $[-L, L]$, with $n = 401$. Put $S = T_w - \Pi$. The enclosed finite-section value is

$$\lambda_{\max}(P_n S P_n) = 1.501365606961153 \dots$$

Together with the outward surrogate and arithmetic radii,

$$\lambda_{\max}(P_n S P_n) \leq 1.5013673873.$$

Hence the direct interval lower bound before the block remainder is

$$\kappa_w^{\text{lo}} - c_L^{\text{hi}} - \lambda_{\max}^{\text{hi}}(P_n S P_n) = 4.2357418881 \cdot 10^{-3}.$$

4.1 Interval construction

The square integral is changed to difference and overlap variables:

$$(T_w)_{ij} = \frac{1}{2} \int_0^{2L} w(x)k(x) \int_{-L}^{L-x} (\phi_i(y+x)\phi_j(y) + \phi_i(y)\phi_j(y+x)) dy dx.$$

Each of $[x_0, x_1]$ and $[x_1, 2L]$ receives a degree-48 Chebyshev surrogate. Bernstein ellipses with respectively $(\rho, M) = (2, 15000)$ and $(1.5, 220)$ give a full-kernel Hilbert–Schmidt surrogate error

$$\varepsilon_{\text{sur}} \leq 1.770301 \cdot 10^{-6}.$$

For the surrogate the inner and outer Gauss–Legendre rules are polynomial-exact. Repeating with twelve extra nodes in both directions changes the matrix by only $2.0894 \cdot 10^{-11}$ in Frobenius norm and $6.5745 \cdot 10^{-12}$ in operator norm; the sealed outward pads are $5 \cdot 10^{-9}$ and 10^{-8} .

The exact Hilbert–Schmidt identity is

$$r_n^2 = \|T_w\|_{\text{HS}}^2 - \sum_{i,j < n} |(T_w)_{ij}|^2, \quad \|T_w\|_{\text{HS}}^2 = \frac{1}{2} \int_0^{2L} w(x)^2 k(x)^2 (2L - x) dx.$$

Projection contraction transfers the surrogate enclosure to this exact expression. Its central value is $6.7727140 \cdot 10^{-4}$ and the certified upper bound is

$$r_{401} \leq 7.0457565325 \cdot 10^{-4} < \frac{1}{3}(4.2357418881 \cdot 10^{-3}).$$

Thus the hard $n \leq 800$ NO-GO does not fire.

4.2 The required threefold charge

For any self-adjoint S ,

$$\lambda_{\max}(S) \leq \lambda_{\max}(P_n S P_n) + 2\|P_n S(1 - P_n)\| + \|(1 - P_n)S(1 - P_n)\|.$$

For T_w , each of the three remainder norms is at most r_n . The two off-diagonal blocks are therefore charged, not discarded:

$$r_n + 2r_n = 3r_n \leq 2.1137269598 \cdot 10^{-3}.$$

The pole operator is rank two. Taylor’s remainder for $\cosh(x/2)$ and $\sinh(x/2)$ after degree 400 bounds its complete block remainder by 10^{-30} . Consequently

$$Q_W^{(w)} \succeq (4.2357418881 - 2.1137269598)10^{-3}I = 2.1220149285 \cdot 10^{-3}I \succeq 2.1 \cdot 10^{-3}I.$$

Together with $Q_W \succeq Q_W^{(w)}$, this proves Theorem 1.

5 Budget and Galerkin audit

The requested human-readable audit decomposition is

$$\underbrace{7.5719610133}_{\text{uncut Galerkin diagnostic}} - \underbrace{3.3344388074}_{\text{booked Loewner loss}} - \underbrace{2.1137269598}_{\text{finite-section block rest}} = 2.1237952461 \quad (\times 10^{-3}).$$

It explains the measured budget but is not the logical lower-bound chain, because the first term is a Galerkin upper bound. The load-bearing operator chain is the direct interval balance in the preceding section and yields $2.1220149285 \cdot 10^{-3}$.

N	6	10	14	18	24	36
$\lambda_{\min}^{\text{even}} \cdot 10^3$	7.596752	7.580831	7.575943	7.573987	7.572755	7.571961

The sequence is monotone from above; its spread from $N = 6$ to $N = 36$ is $2.4791 \cdot 10^{-5}$. Every entry remains explicitly typed as a finite-section upper bound.

6 Kill and pad audit

Consolidated family	R494 disposition
F-I high-band majorants	absent: the proof uses positive translation differences and one compact kernel
F-II support-mismatched bases	absent: Legendre functions live on the support interval and only certify a finite section
F-III unresolved discrete modes	absent: the exact complementary block is enclosed by r_{401}
F-IV quadrature/truncation pads	absent: the cutoff loss is booked and the surrogate has an outward Bernstein enclosure
F-V wrong comparison objects	absent: no shifted comparison, Schur pad, or asymmetric product enters
F-VI false operator premises	absent: no inverse or coercivity premise is used
N1	blocked: Galerkin numbers are labelled upper bounds only
N2	blocked: $[0, x_0]$ is booked through κ_w and the loss line
N3	blocked: the rest is $r + 2r = 3r$, including both off-block terms
N4	blocked: after the mandatory G1 identity calibration, the proof stays entirely in the x -space translation/kernel representation

7 Scaling beyond the prime-free rung

When $2L \geq \log 2$, the prime term is no longer zero. It enters the same representation as the explicit bounded translation sum

$$- \sum_{n \leq e^{2L}} \frac{\Lambda(n)}{\sqrt{n}} (R_{\log n} + R_{-\log n}).$$

At $L = 0.8$ there are exactly three terms, $n = 2, 3, 4$. Each compressed translation has norm at most one, so this is a finite bounded perturbation. The Loewner monotonicity of the positive archimedean translation-difference integral is unchanged. Unlike the present $L = 0.3$ rung, the literal nodes $\log n$ make the scramble gate live there.

8 What is not claimed

No Galerkin eigenvalue is promoted to an operator value. No result for every support length is asserted. No zero statistic or zero oracle is used. The theorem is an unconditional fixed-support Weil-positivity statement in the classical Yoshida–Bombieri zone, not an RH claim and not an anti-RH claim.